Department of Computer Science
National University of Computer & Emerging Sciences
Chiniot-Faisalabad

MT1003-Calculus & Analytical Geometry

Fall 2022

Assignment # 4

DUE DATE: 30 November 2022

Instructions for submission of assignment:

- Assignment should be on **A4** paper stapled in the left upper corner.
- At the top of the first page, you must put
  - On the left your name with your Roll number.
  - On the right your section.
- No assignment will be submitted after the due date.
- If found with any plagiarism, the assignment submission will be canceled. An individual **VIVA EXAM** of each student based particularly on this Assignment may be conducted to establish self-attempt of this Assignment.

Question # 1

Find the slopes of the tangent lines to the curve $y^2 - x + 1 = 0$ at the points $(2, -1)$ and $(2, 1)$.

Question # 2

Use implicit differentiation to find d^2y/dx^2 if

i) $4x^2 - 2y^2 = 9$

ii) $\dfrac{xy^3}{1 + \sec y} = 1 + y^4$

Question # 3

Show that

$$f(x) = \begin{cases} x^2 + 1, & x \le 1 \\ 2x, & x > 1 \end{cases}$$

is continuous and differentiable at $x = 1$. Sketch the graph of f.

Question # 4

$$\lim_{x\to 1^+}\left[(x-1)\tan\left(\frac{\pi}{2}x\right)\right]$$

Question # 5

$$\lim_{y\to 0^+}[\cos(2y)]^{1/y^2}$$

Question # 6

$$\lim_{x\to \pi}\frac{\sqrt{1-\tan x}-\sqrt{1+\tan x}}{\sin 2x}$$

Question # 7

$$\lim_{x\to 1}\frac{\tan^{-1}x-\frac{\pi}{4}}{\tan\frac{\pi}{4}x-1}$$

Question # 8

$$\lim_{x\to -1}\frac{\sqrt{x+10}+3x^{1/3}}{4x^2+3x-1}$$

Question # 9:

$$\lim_{x\to 0}\left(\frac{1}{\sin^2 x}-\frac{1}{x^2}\right)$$

Question # 10:

$$\lim_{x\to 0}\left(\frac{e^x}{e^x-1}-\frac{1}{x}\right)$$

Question # 11

Find the derivative of $y=3\log_7(x^2+1)$.

Question # 12

. Find the derivative of

$$3 \ln xy + \sin y = x^2.$$

Question # 13

Use Rolle's Theorem for the given function and interval.

$g(t) = 2t - t^2 - t^3$ on $[-2, 1]$

Question # 14

Use Mean Value Theorem for the given function and interval.

$h(z) = 4z^3 - 8z^2 + 7z - 2$ on $[2, 5]$

Question # 15

Differentiate w.r.t x,

$$y = \ln(\ln(\ln x))$$

Question # 16

Differentiate w.r.t x,

$$y = x[\log_2(x^2 - 2x)]^3$$

Question # 17

Differentiate w.r.t x,

. $\frac{d}{dx}[\ln((x-1)^3(x^2+1)^4)]$

Question # 18

Differentiate w.r.t x,

$$\frac{d}{dx}\left[\ln \frac{\cos x}{\sqrt{4-3x^2}}\right]$$

Question # 19:

Given the function $f(x) = e^{3x+2} - 3x$

a) Find all critical points.
b) List the intervals in which f(x) is increasing.
c) List the intervals in which f(x) is increasing.
d) List all of the points at which f(x) has a local maximum.
e) List all of the points at which f(x) has a local minimum.

Question # 20:

Given the function $f(x) = 3 - (2x+5)^{2/3}$

a) Find all critical points.
b) List the intervals in which f(x) is increasing.
c) List the intervals in which f(x) is increasing.
d) List all of the points at which f(x) has a local maximum.
e) List all of the points at which f(x) has a local minimum.

Question # 21

Find all local extrema for $f(x) = 2x^3 - 9x^2 - 24x + 11$

Question # 22:
Given function is,

$$h(t) = t^4 + 12t^3 + 6t^2 - 36t + 2$$

(a) Determine a list of possible inflection points for the function.

(b) Determine the intervals on which the function is concave up and concave down.

(c) Determine the inflection points of the function.

Question # 23: **Given function is,**

$$h(w) = 8 - 5w + 2w^2 - \cos(3w) \text{ on } [-1, 2]$$

(a) Determine a list of possible inflection points for the function.

(b) Determine the intervals on which the function is concave up and concave down.

(c) Determine the inflection points of the function.

Question # 24

$P(w) = w\mathbf{e}^{4w}$ on $\left[-2, \frac{1}{4}\right]$

(a) Identify the critical points of the function.

(b) Determine the intervals on which the function increases and decreases.

(c) Classify the critical points as relative maximums, relative minimums or neither.

(d) Determine the intervals on which the function is concave up and concave down.

(e) Determine the inflection points of the function.

(f) Use the information from steps **(a)** – **(e)** to sketch the graph of the function.

GOOD LUCK 😊